Prop. Normal and Nilpotent $\Rightarrow$ Zero.

Proof. Let $T: V \to V$ be a linear operator on fin. dim inn. prod space V. Since T is normal, there exists a basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ for V consisted by eigenvectors of T. Given $\alpha \in V$,

$$\alpha = x_1 \alpha_1 + \cdots + x_n \alpha_n \quad \text{for some } x_i \in F$$

and so $T\alpha = x_1 c_1 \alpha_1 + \cdots + x_n c_n \alpha_n$, where c_i are eigenvalues corresponding to α_i.
Meanwhile, since T is nilpotent, all eigenvalues for T are zero, so $T\alpha = 0$.
Thus, $T \equiv 0$.

Alternatively, without spectral decomposition, observe that: if T is normal,

$$\|T\alpha\|^2 = \langle T\alpha, T\alpha \rangle = \langle \alpha, T^* T\alpha \rangle = \langle \alpha, T T^* \alpha \rangle = \langle T^*\alpha, T^*\alpha \rangle = \|T^*\alpha\|^2$$

so $T\alpha = 0 \iff T^*\alpha = 0$, which gives $\ker T = \ker T^*$.
Now, suppose that T is nilpotent and $T \neq 0$. Take $m \in \mathbb{N}$ be the smallest integer such that $T^m \equiv 0$. (That is, $T^0, \cdots, T^{m-1} \neq 0$).
Let $\alpha \in V$ such that $T^{m-1}\alpha \neq 0$. Then, since $T^m = 0$, $T^{m-1}\alpha \in \ker T$.
Since $\ker T = \ker T^*$, $T^* T^{m-1}\alpha = 0$. So,

$\langle T^* T^{m-1}\alpha,$

「Prop.

Let T be a normal operator on a fin. dim complex inn. prod space V.
Then, T^* is polynomial in T.